

EXPLORING CUSTOMER BEHAVIOR THROUGH DATABASE ANALYSIS

A Real-Life Data Analyst
Case Study



Introduction



- In today's competitive online food delivery industry, understanding customer behavior is key to improving sales, customer loyalty, and overall business growth.
- This project is a **case study** based on an online food ordering platform. The dataset includes **5 tables**:
 - **Sales** (orders placed by customers)
 - **Product** (menu items with prices)
 - **Users** (customer signup details)
 - **User_Name** (user IDs with names)
 - **Goldusers_Signup** (customers who upgraded to gold membership)

Introduction



- The main objective is to analyze customer purchase patterns, product popularity, and gold membership performance using SQL queries.
- By uncovering these insights, the company can make **data-driven decisions** in :
 - Expanding its loyalty program.
 - Optimizing product offerings.
 - Improving customer experience through personalization .

Methodology

Data and Tools

- **Data Source:** Customer transactions dataset (5 relational tables)
- **Database Design:** Defined primary and foreign keys.
- **Tools:** Sql Server was used for querying.



Methodology

Process Flow

- **Step1:** Created the 5 tables and inserted data into it.
- **Step2:** Ran queries to get insights about the businesss.
- **Step3:** Used **JOIN, GROUP BY, SUM, COUNT, RANK, DATEDIFF** for deeper insights
- **Step4:** Compared **Gold vs Non-Gold users** in terms of revenue & behavior.
- **Step5:** Derived trends → yearly sales, product popularity, customer preferences.



Key Findings

Sales & Product

- **Top Revenue Product:** Butter Chicken with Rs.1700 total revenue
- **Top 3 Products by Revenue:**
 - Butter Chicken – ₹1700
 - Shahi Paneer – ₹1360
 - Mutton Biryani – ₹1350
- **Lowest Revenue Product:** Mango Lassi with ₹80 revenue
- **Most Purchased Item:** Shahi Paneer with highest order count



Key Findings

Gold vs Non-Gold users

- There are **total of 10 users** out of which **5 are gold users**.
- **Revenue Contribution:**
 - Gold Users– ₹3830(55%)
 - Non-gold Users – ₹3060(45%)
- **Popular Product Among Gold Users:** Dal Makhani
- **Gold membership Tenure:**
 - Anshul & Sahil – 8 years
 - Sreya & Aryan– 7 years
 - Priya– 6 years



Key Findings

Customer Trends

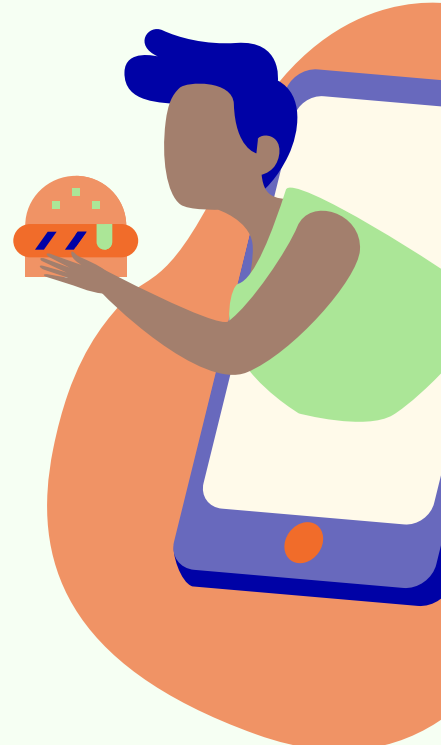
- **Yearly Revenue Trends:**
Highest Revenue in 2017 and Lowest Revenue in 2014 .
Growth peaked 2016 → 2017, slight decline afterward.
- **First Orders:**
Dal Makhani was the first order for 4 users.
Chicken Tikka was the first order of 2 users.
- **Behaviour Insights:**
Gold members place more frequent orders
Spending before & after gold signup varies, showing increased loyalty post-membership
Clear favorites suggest scope for personalized recommendations



Insights

Customer Behaviour

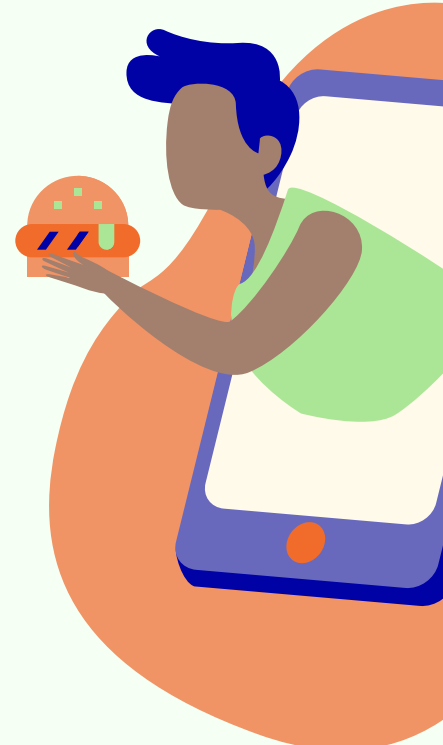
- Customers show **strong loyalty to specific dishes** like Butter Chicken, Shahi Paneer and Dal Makhani
- **Gold members consistently spend more** and order more frequently proving the effectiveness of the loyalty program.
- **Non-gold users contribute less revenue**, highlighting the importance of converting them.
- First orders (like Dal Makhani, Chicken Tikka) often become **gateway dishes** that drive repeat behavior.



Insights

Sales & Trends

- **Sales revenue peaked in 2017** but declined afterward → indicates potential customer churn or lack of new engagement strategies.
- Certain products (Mango Lassi, Veg Curry) remain **underutilized**, suggesting mismatched customer preferences.
- **Gold membership signup dates spread over years (2017–2019)** → shows steady adoption but also room to accelerate growth.
- Seasonal or yearly variations show that **marketing campaigns should align with peak demand periods**.



Actionable Insights

Product & Revenue

- **Promote Top Products:** Butter Chicken, Shahi Paneer, and Mutton Biryani are revenue leaders → highlight them in ads & special combos.
- **Boost low selling item:** Mango Lassi & Veg Curry have very low sales → either improve visibility or phase them out.
- **Personalized Recommendations:** Each customer has a clear favorite dish → recommend these items in repeat orders to increase loyalty.



Actionable Insights

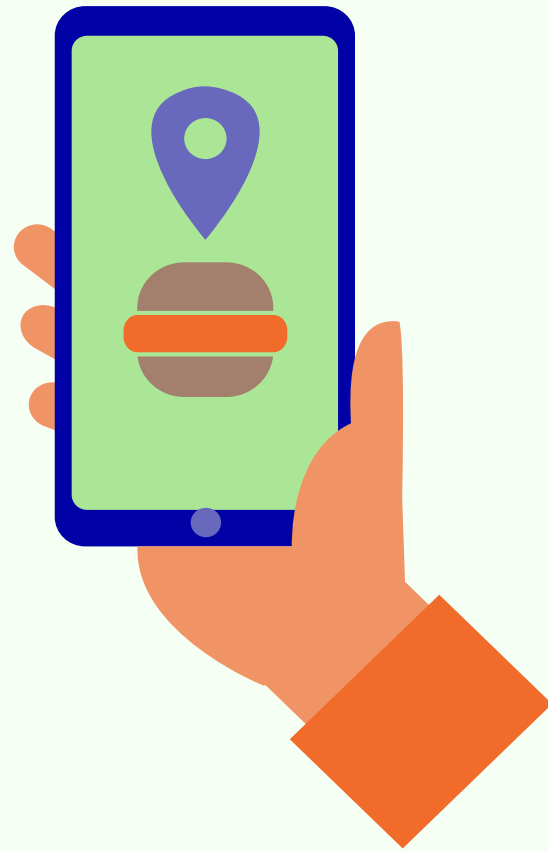
Gold Membership and Customers

- **Expand gold membership:** Gold users spend 55% of total revenue despite being only 50% of users → push upgrade offers to non-gold customers
- **Retention Strategy:** Revenue peaked in 2017, then declined → re-engage inactive customers with discounts, festive offers, or loyalty points.
- **Personalized Recommendations:** Each customer has a clear favorite dish → recommend these items in repeat orders to increase loyalty.
- **Leverage first order Insights:** Dal Makhani is the most common first dish ordered → offer it as a “Welcome Dish Discount” to attract new users.



Conclusion

- This project analyzed **customer behavior** using SQL queries on sales, users, products, and gold membership data.
- Analysis of customer sales data revealed **clear purchase patterns**.
- **Gold Members** spend more & order more → program adds strong value.
- Revenue peaked in 2017, slight decline afterward.
- Customers show **loyalty to favorites** & repeat behaviors.
- Findings highlight the need for **targeted promotions, stronger engagement, and gold membership expansion**.





Thank You!